

Common setup — Wenbin UQ (frozen)

research/formal (working)

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1 Objects

Let $\mathcal{D} = \{(X_i, Y_i)\}_{i=1}^n$ denote calibration data. Let θ be a hyperparameter of an uncertainty set $\mathcal{C}(X; \mathcal{D}, \theta)$. Miscoverage level $\alpha \in (0, 1)$.

Assumption 1 (Fixed- θ validity). *For any given $\mathcal{D}$ and prespecified θ ,*

$$\mathbb{P}(Y \in \mathcal{C}(X; \mathcal{D}, \theta)) \geq 1 - \alpha. \quad (1)$$

Assumption 2 (Data-dependent selection program). *There exist $\hat{f}, \hat{g}$ with $\sigma(\hat{f}), \sigma(\hat{g}) \subseteq \sigma(\mathcal{D})$ and*

$$\hat{\theta}(\mathcal{D}) \in \arg \min_{\theta} \{\hat{f}(y; \theta) : \hat{g}(\theta) \leq 0\}. \quad (2)$$

Remark 1 (Notation quarantine). *Writeup (1) uses third argument θ ; prose may say α . Writeup uses lowercase y in $\hat{f}(y; \theta)$. See `research/notes/notation_quarantine.md`. Local convention here: treat y in (2) as a placeholder argument of $\hat{f}$ (possibly a fixed template outcome / empirical functional of $\mathcal{D}$), not necessarily equal to the test Y in (1). Flag if lemmas need a sharper reading.*

Remark 2 (Alias $C^{(q)}$). *Later Part II files may write $C^{(q)}(X; \mathcal{D}, \theta)$ for the writeup set $\mathcal{C}(X; \mathcal{D}, \theta)$ run at classical miscoverage level q (instead of α). This does **not** put the level into the third slot; θ remains the optimized hyperparameter.*

Definition 1 (Leave-one-out neighbor). $\mathcal{D}^{(i)}$ *is obtained from $\mathcal{D}$ by replacing the i -th pair by an independent draw from the same distribution (or deleting it — state which when used).*

Definition 2 (Selected object). *The live selected object is only $\hat{\theta}(\mathcal{D})$ from (2).*

2 Goals (reference)

- (i) When does (1) hold at $\theta = \hat{\theta}(\mathcal{D})$?
- (ii) If not, what is the nominal-vs-actual miscoverage deviation?
- (iii) Restore (1) at the selected hyperparameter *without* recalibrating $\mathcal{C}$, by data-randomizing the selection map (W5).